Fraud Ring Radar

Detecting Coordinated Fraud Networks with Multi-Agent AI, Graph Memory, and Bayesian Confidence

TEAM FRAUDKILLER

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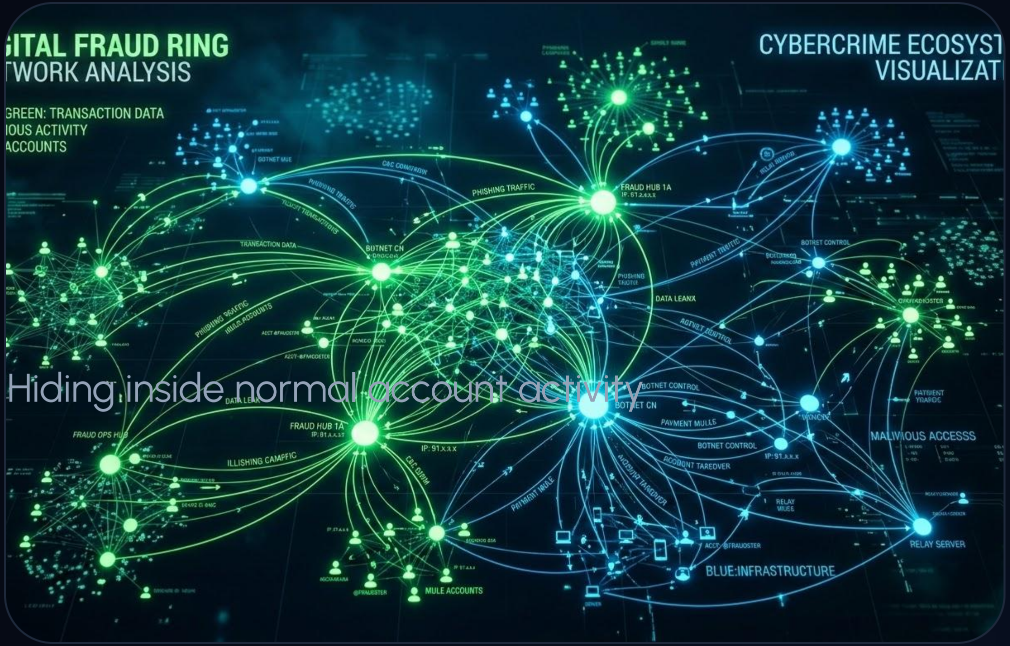
Why Traditional Fraud Systems Miss Coordinated Fraud

Most fraud monitoring systems analyze transactions **one row at a time**.

Fraud rings exploit this by:

- Staying below alert thresholds
- Splitting transfers into many small payments
- Coordinating across multiple accounts
- Sharing devices and infrastructure
- Hiding inside normal transaction activity



`The fraud is not in a single transaction.`

`The fraud is in the graph.`

Inspiration & Vision

Built for Analysts Who Only Have Minutes

We wanted to build a tool that helps fraud investigators:

- Find the suspicious ring
- Understand why it matters
- Measure confidence, not just suspicion
- Generate investigator-ready summaries

Goal: **Transform hours of investigation into minutes.**

Live Demo Flow

Ingest

Upload raw transaction data and device logs.

Map

Build graph memory connecting IPs and merchants.

Detect

Detect coordinated loops and smurfing clusters.

Rank

Rank cases by Bayesian confidence scores.

Co-Pilot

Ask the AI questions about case connections.

Export

Generate investigator narratives and SAR reports.

Multi-Agent Pipeline

1 GraphRAG Builder

Ingests data via **Cognee** to connect accounts, devices, and IPs.

2 Cluster Detector

Identifies circular money loops, smurfing, and merchant laundering hubs.

3 Case Summarizer

Ranks findings and produces structured SAR-style evidence reports.

Seeing What Rules Miss

Traditional Rule Engine

Transaction → Threshold → Alert

Misses rings staying below limits or splitting transfers.

Fraud Ring Radar

Graph → Relationships → Intelligence

Uncovers coordinated timing, routing hubs, and shared infrastructure.

Benchmark Result: The fraud ring remained below transaction thresholds but became obvious once viewed as a network.

Bayesian Confidence Layer

Agents find the behavior; **PyMC** quantifies the certainty.

$$P_{\text{Fraud}} = \sum \left(E_{\text{Graph}} + E_{\text{Behavior}} + E_{\text{Infra}} \right)$$

Evidence Source	Signal Captured	Model Weighted Impact
Graph Topology	Circular loops & Cluster size	High (Structural certainty)
Behavioral Logs	Night-time bursts & reactivation	Medium (Temporal anomaly)
Infrastructure	Shared device IDs & IP clusters	Very High (Hard linkage)

From Detection to Decision

STEP 01	STEP 02	STEP 03	STEP 04	STEP 05
Data Ingest	**Agent Scan**	**Graph Evidence**	**PyMC Scoring**	**Action**
Raw Tx & Logs	Multi-agent detection	Extracted Features	Probability + Interval	Escalate/Block

PyMC Confidence Output:

96% (Interval: 91-99%)

⚠ **RECOMMENDED ACTION: ESCALATE**

Exposure: $161,751 | Members: 12

Quantifying Business Impact

383
SUSPICIOUS FRAUD CASES

284
HIGH RISK ACCOUNTS

$156K+
POTENTIAL EXPOSURE UNCOVERED

Efficiency Gain

Complete case review in under **5 minutes**.

Traditional investigation would require hours of manual CSV filtering and visualization to trace these same loops.

Go-To-Market Strategy

Target Customers

- Community Banks
- Credit Unions
- Mid-Market FinTechs

Privacy-First Positioning

Local deployment available. No core banking replacement required. Explainable investigations for compliance.

Business Model

Community Edition	Open-Source Prototype
Enterprise SaaS	Cloud-Managed Platform
On-Premise	Privacy-Sensitive Deployment
Add-Ons	Real-time Ingestion & Compliance

The Roadmap Ahead

Real-time Kafka Ingestion
From batch processing to sub-second detection swarms.

Identity Resolution
Advanced entity linking across multiple data silos.

SAR Automation
One-click export for regulatory filing (FinCEN compliant).

Continuous Learning
Bayesian priors that update as analysts confirm or reject alerts.

Core Integrations
Native connectors for Fiserv, Jack Henry, and modern BaaS.

Audit Transparency
Full history of agent actions for regulatory reviews.

Questions?

Agents discover. PyMC quantifies. Analysts decide.

Thank you for your time.

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